



Unit – 3

Combinatorial Mathematics

By Prof. Jasleen Kaur
Department of Information Technology
Guru Nank Dev Engineering College Ludhiana

OUTLINE

1. Introduction
2. Basic Counting Principles (sum rule, product rule)
3. Permutations and Combinations
4. Pigeonhole Principle
5. Recurrence Relations : Solving Homogeneous recurrence relations and Non-Homogeneous recurrence relations.
6. Generating function

Introduction

- ▶ **Combinatorial mathematics** (Combinatorics) is the branch of mathematics focused on counting, arranging and analyzing finite or discrete structures.
- It involves determining the number of possibilities without explicitly enumerating them using techniques like permutations, combinations and the principle of inclusion-exclusion.
- Key applications include computer science, cryptography, optimization and probability
- Two most basic rules:-
 - Sum rule (Addition Rule)
 - Product rule (Multiplication Rule)
- Combining the product rule and the sum rule. Thus we can solve more interesting and complex problems



Basic Counting Principles

- ▶ **Addition Rule** states that for two possible events A and B where A and B both are mutually exclusive events, i.e. they have no outcome in common (disjoint) and if event E is defined as occurring in either event A **OR** event B then the possible number of ways in which event E can occur is,

$$n(E) = n(A \cup B) = n(A) + n(B)$$

- ▶ The number of ways that “**either task 1 or task 2 can be done, but not both**” is $m + n$.
- ▶ Example 1: A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects respectively. How many possible projects are there to choose from?

Solution: $23+15+19$

- ▶ **Multiplication Rule** states that for "n" mutually independent simultaneous events, P1, P2, P3, ...Pn. Number of ways in which these events can occur is n(P1), n(P2), n(P3),... n(Pn) respectively. Now we define an event E such that it is happening all the events simultaneously then the number of ways this can happen is,

$$n(E) = n(P1) \times n(P2) \dots \times n(Pn)$$

- ▶ The number of ways that “**both tasks 1 and 2 can be done**” in *mn*.
- ▶ Example 1: How many different 7-digit phone numbers can be formed if the first digit cannot be 0 or 1 and no digit can be repeated?

Solution: First digit: 8 choices (2 - 9), Second digit: 9 choices (any except the first) Third digit: 8 choices, Fourth digit: 7 choices, Fifth digit: 6 choices, Sixth digit: 5 choices, Seventh digit: 4 choices.

Total = $8 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 = 483,840$ numbers

Example 3: How many different ways can the letters of the word "STATISTICS" be arranged if all the S's must be together and all the T's must be together?

Solution:

Treat all S's as one unit and all T's as one unit

We now have 6 units to arrange: SSS, TTT, A, I, I, C

This is a permutation with repetition (2 I's)

Number of arrangements = $6! / 2! = 360$

The S's can be arranged internally in $3!$ ways but they are identical so $3!/3!=1$

Similarly, The T's can be arranged internally in $3!$ ways but they are identical so $3!/3!=1$

Total arrangements = 360

The Pigeonhole Principle

- The Pigeonhole Principle states that if items(pigeons) are put into containers(holes), then at least one container must contain more than one item.
- If $k+1$ objects are assigned to k places, then **at least** one place must be assigned ≥ 2 objects.
- This fundamental counting technique, often used in discrete mathematics and computer science, guarantees that if you have more items than slots, at least one slot will have a collision.



Example 1: If 5 pigeons are placed into 4 holes. How many holes will contain 2 or more pigeons?

Solution: At least one hole must contain 2 or more pigeons.

Importance of Pigeonhole Principle:

- **Helps in proving existence:** It is used to prove that at least one repeated or common element must exist in a set of objects.
- **Helps in detecting repetition or duplication:** It guarantees that repetition must occur when the number of objects is greater than the number of categories.
- **Useful in solving counting problems:** It is an important tool in Combinatorics for solving problems related to arrangements and distributions.
- **Simplifies mathematical proofs:** Many mathematical results can be proved easily using this principle without complex calculations.
- **Applications in computer science:** It is used in Computer Science for problems related to hashing, data storage and algorithm design.
- **Used in graph theory and number theory:** The principle helps in solving problems in Graph Theory and Number Theory.
- **Widely used in logical reasoning and puzzles:** Many logical puzzles and mathematical problems are based on this principle.

Generalized Pigeonhole Principle: If n pigeonholes are occupied by $kn + 1$ or more pigeons, where k is a positive integer, then at least one pigeonhole is occupied by $k + 1$ or more pigeons.

Example 1: Find the minimum number of students in a class to be sure that three of them are born in the same month.

Solution: Here the $n = 12$ months are the pigeonholes, and $k + 1 = 3$ so $k = 2$. Hence among any $kn + 1 = 25$ students (pigeons), three of them are born in the same month.

Example 2: If there are 280 people in party. How many Students at least can have same birthday in a Week.

Solution: There are $N = 280$ people in a party. There are $k = 52$ weeks in year.

Therefore, there must be at **least** 1 week during which **at least** $\lceil 280/52 \rceil = \lceil 5.38 \rceil = 6$ students in the party have a birthday.

Example 3: If there are 100 students, how many students possibly share same birth month?

Solution: 100 students and 12 months:- $100/12 = 9$. Therefore, at least 9 students share a birth month.

Example 4: Suppose a department contains 13 professors, then two of the professors (pigeons) were born in the same month (pigeonholes)

Example 5: Find the minimum number of elements that one needs to take from the set $S = \{1, 2, 3, \dots, 9\}$ to be sure that two of the numbers add up to 10. Here the pigeonholes are the five sets $\{1, 9\}$, $\{2, 8\}$, $\{3, 7\}$, $\{4, 6\}$, $\{5, 5\}$.

Thus any choice of six elements (pigeons) of S will guarantee that two of the numbers add up to ten.

Recurrence Relation

- ▶ A **recurrence relation** is a mathematical expression that defines a sequence in terms of its previous terms.

- ▶ General form of a **Recurrence Relation**:

$$a_n = f(a_{n-1}, a_{n-2}, \dots + a_{n-k})$$

where **f** is a function that defines the relationship between the current term and the previous terms.

- Example:

- Fibonacci Sequence: $F(n) = F(n-1) + F(n-2)$
- Factorial $F(n) = n * F(n-1)$
- Ackermann function

- **Types:**

- Homogeneous Recurrence Relations
- Non-Homogeneous Recurrence Relations

A recurrence relation of the form : $\mathbf{a_n = ca_{n-1} + f(n)}$ for $n \geq 1$ where c is a constant and $f(n)$ is a known function is called linear recurrence relation of first order with constant coefficient. If $f(n) = 0$, the relation is homogeneous otherwise non-homogeneous.

Please see class notes!!!!

Generating Functions

- In mathematics, a generating function is a representation of an infinite sequence of numbers as the coefficients of a formal power series.

Generating Functions

$$G(x) = \sum_{k=0}^{\infty} a_k x^k$$

- The generating function are single function which encodes the sequence.
- Instead of an infinite sequence (for example: 1,3,5,7,9,...)
- Generating functions are sometimes called Generating series, in that a series of terms can be said to be the generator of its sequence of term coefficients.

An infinite power series is simply an infinite sum of terms of the form $c_n x^n$ where c_n is some constant. So we might write a power series like this:

$$\sum_{k=0}^{\infty} c_k x^k.$$

or expanded like this

$$c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + \dots$$

Questions For Practice:

- **Ques1:** Find the generating function for the sequence $\{1,2,3,4,5,\dots\}$
- **Ques2:** Find the generating function for the sequence $\{2,4,6,8,10,\dots\}$
- **Ques3:** Derive the generating function for the sequence $\{3,6,9,12,\dots\}$

- **Please see class notes!!!!**

Ackermann Function

$$A(m,n) = \begin{cases} n + 1 & \text{if } m=0 \\ A(m-1,1) & \text{if } m > 0 \text{ and } n = 0 \\ A(m-1,A(m,n-1)) & \text{if } m > 0 \text{ and } n > 0 \end{cases}$$

Question: Solve $A(1, 2)$?

Solution: Given problem is $A(1, 2)$ Here $m = 1, n = 2$

where m and n are non-negative integers

As $m > 0$ and $n > 0$, Hence apply third condition of Ackermann function

$A(1, 2) = A(0, A(1, 1))$ ----- (1) // Now, to find $A(1, 1)$ apply third condition

$A(1, 1) = A(0, A(1, 0))$ ----- (2) // Now, to find $A(1, 0)$ apply second condition

$A(1, 0) = A(0, 1)$ ----- (3) // Now, to find $A(0, 1)$ apply first condition

$$A(0, 1) = 1 + 1 = 2$$

Now put this value in equation 3, Hence $A(1, 0) = 2$

Now put this value in equation 2, $A(1, 1) = A(0, 2)$

[find $A(0, 2)$ by applying first condition as $A(0, 2) = 2 + 1 = 3$]

Hence $A(1, 1) = 3$

Now put this value in equation 1 $A(1, 2) = A(0, 3)$

[find $A(0, 3)$ by applying first condition as $A(0, 3) = 3 + 1 = 4$]

Hence $A(1, 2) = 4$

So, $A(1, 2) = 4$

Questions For Practice:

Question: Solve $A(2, 1)$?

Answer: 5

Question: Solve $A(2, 2)$?

Answer: 7

Question: Solve $A(1, 3)$?

Answer: 5

THANKS